IGCSE 0607 Practice Questions — E8.2 — Combined Questions & Markscheme

Vectors in two dimensions

Total Questions: 21 — **Total Marks:** 132 Marks

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1 Vectors $\mathbf{p}$ and $\mathbf{q}$ are given as

$$\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \qquad \mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$$

(a) Work out

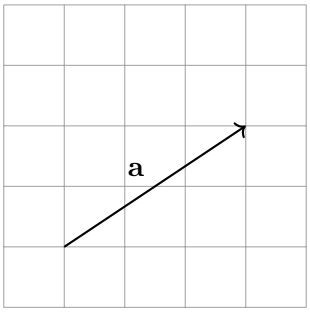
$$2\mathbf{p} + \mathbf{q}$$

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Point A has coordinates $(1, 4)$.
The translation from A to B is given by the vector $\mathbf{p}$.
Work out the coordinates of B.

(..... , ...) **[1]**

2 The diagram shows a vector **a**.



(a) Write **a** as a column vector.

$\mathbf{a} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

(b) $\mathbf{b} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.
Work out $2\mathbf{a} - \mathbf{b}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

3 The vectors **a** and **b** are given by

$\mathrm{a} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ and $\mathrm{b} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$ (0)

(a) Work out $\mathbf{a} + 2\mathbf{b}$ as a single column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Point P has coordinates $(3, -1)$.
Point Q is translated from P by the vector **a**.
Find the coordinates of point Q.

(..... , ...) **[2]**

4 Given the vectors $\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$.

(a) Work out $3\mathbf{p}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

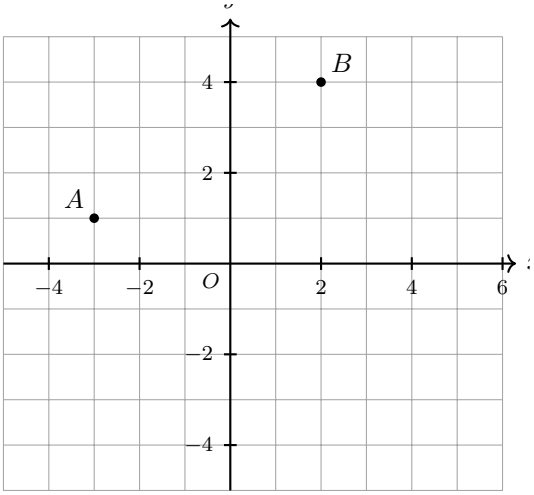
(b) Work out $\mathbf{p} - 2\mathbf{q}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(c) Point M is translated to point N by the vector $\mathbf{q}$.
Given that the coordinates of N are $(6, -1)$, find the coordinates of M.

(..... , ...) **[2]**

5 The diagram shows points $A(-3, 1)$ and $B(2, 4)$ on a coordinate grid.



(a) Write down $\vec{AB}$ as a column vector.

$\vec{AB} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) $\mathbf{p} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.

Calculate

(i) $\mathbf{p} + 2\mathbf{q}$,

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

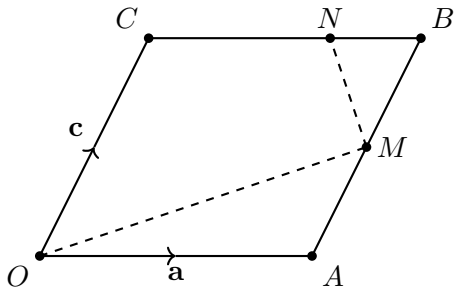
(ii) $3\mathbf{p} - \mathbf{q}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(c) Point E is such that $\vec{BE} = \begin{pmatrix} -4 \\ -6 \end{pmatrix}$.
Find the coordinates of E.

$E = (\ldots\ldots\ldots \quad , \quad \ldots\ldots\ldots)$ **[2]**

6 The diagram shows a parallelogram $OABC$, where O is the origin.
$\vec{OA} = \mathbf{a}$ and $\vec{OC} = \mathbf{c}$.
M is the midpoint of AB and N is the point on BC such that $BN : NC = 2 : 1$.



(a) Find each of the following vectors in terms of **a** and **c**.
Give each answer in its simplest form.

(i) $\vec{OM}$

$\vec{OM} =$ ..... **[2]**

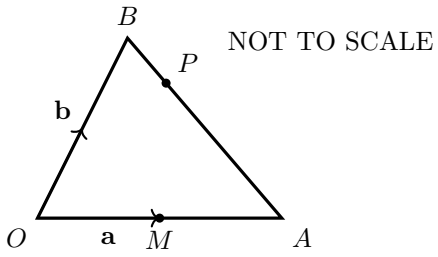
(ii) $\vec{MN}$

$\vec{MN} =$ ..... **[2]**

(b) $\vec{ON} = k\mathbf{a} + \mathbf{c}$.
Write down the value of k.

$k =$ ..... **[2]**

7 The diagram shows triangle OAB with $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$.
M is the midpoint of OA, and P is the point on AB such that $AP : PB = 1 : 3$.



(a) Express each of the following vectors in terms of **a** and **b**, giving your answers in their simplest form:

(i) $\vec{AB}$,

$\vec{AB} =$ ..... **[1]**

(ii) $\vec{OP}$,

$\vec{OP} =$ ..... **[2]**

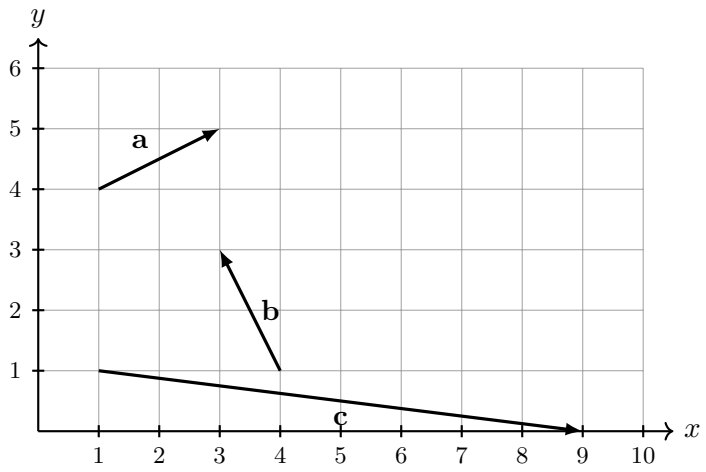
(iii) $\vec{MP}$.

$\vec{MP} =$ ..... **[2]**

(b) The line OP is extended to meet the line through B parallel to OA at the point C, where $\vec{BC} = k\mathbf{a}$ and $\vec{OC} = m\vec{OP}$.
Find the value of k and the value of m.

$k =$ ..... , $m =$ ..... **[4]**

8 The diagram shows vectors **a**, **b** and **c** plotted on a Cartesian coordinate grid.



(a) Write **c** as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

(b) Vector **c** can be written in the form $p\mathbf{a} + q\mathbf{b}$, where p and q are integers.
Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[4]**

9 Vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are given by

$$\mathbf{u} = \begin{pmatrix} 3 \\ m \end{pmatrix} \qquad \mathbf{v} = \begin{pmatrix} n \\ -2 \end{pmatrix} \qquad \mathbf{w} = \begin{pmatrix} 10 \\ 7 \end{pmatrix}$$

(a) Given that $2\mathbf{u} - \mathbf{v} = \mathbf{w}$, find the value of m and the value of n.

$m =$ ..... , $n =$ ..... **[3]**

(b) Vector $\mathbf{z} = \begin{pmatrix} k \\ 12 \end{pmatrix}$.
Given that $\mathbf{z}$ is parallel to $\mathbf{u}$, find an expression for k in terms of m.

$k =$ ..... **[3]**

10 $\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ $\mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$

(a) Work out $3\mathbf{p} - 2\mathbf{q}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Point A has coordinates $(3, -1)$.
$\vec{AB} = \mathbf{p} + \mathbf{q}$.
Find the coordinates of B.

(..... , ...) **[2]**

11 Vectors $\mathbf{p}$ and $\mathbf{q}$ are given as:

$$\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \qquad \mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$$

(a) Work out $3\mathbf{p}$.

$3\mathbf{p} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

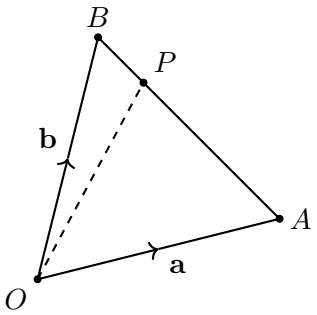
(b) Work out $\mathbf{p} - 2\mathbf{q}$.

$\mathbf{p} - 2\mathbf{q} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(c) Point A has coordinates $(3, -1)$.
Point B is the image of A after a translation by the vector $\mathbf{p}$.
Work out the coordinates of B.

(..... , ...) **[2]**

12 In triangle OAB, $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$.
P is the point on AB such that $AP : PB = 3 : 1$.



(a) Find $\vec{OP}$ in terms of **a** and **b**.
Give your answer in its simplest form.

$\vec{OP} =$ ..... **[3]**

(b) M is the midpoint of OA.
Find $\vec{MP}$ in terms of **a** and **b**.
Give your answer in its simplest form.

$\vec{MP} =$ ..... **[2]**

13 Vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are given by:

$$\mathbf{u} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} \qquad \mathbf{v} = \begin{pmatrix} x \\ 5 \end{pmatrix} \qquad \mathbf{w} = \begin{pmatrix} -1 \\ y \end{pmatrix}$$

(a) Given that $2\mathbf{u} + \mathbf{v} = \mathbf{w}$, find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[3]**

(b) Point M has coordinates $(4, 7)$ and $\vec{MN} = \begin{pmatrix} -2 \\ 6 \end{pmatrix}$.

(i) Find the coordinates of N.

(..... , ...) **[2]**

(ii) Write down the vector $\vec{NM}$.

$\vec{NM} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

14 **(a)** $\mathbf{u} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} -1 \\ y \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 10 \\ 6 \end{pmatrix}$.

Given that $3\mathbf{u} + k\mathbf{v} = \mathbf{w}$, work out the value of k and the value of y.

$k =$ ..... , $y =$ ..... **[3]**

(b) Points P, Q and R have coordinates $(2, -1)$, $(5, 5)$ and $(11, 17)$ respectively.

(i) Write $\vec{PQ}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

(ii) Show that P, Q and R lie on a straight line.

[2]

15 The coordinates of point A are $(2, 3)$ and the coordinates of point B are $(-1, 5)$.

(a) Write down the column vector $\vec{AB}$.

$\vec{AB} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

(b) Point C is such that $\vec{BC} = 3\vec{AB}$.
Find the coordinates of C.

(..... , ...) **[3]**

(c) $\mathbf{p} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$.
Find the column vector $\mathbf{q}$ such that $\vec{AB} + \mathbf{q} = \mathbf{p}$.

$\mathbf{q} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

16 Points A, B and C have coordinates $(1, -2)$, $(7, 10)$ and $(4, 4)$ respectively.

(a) Write $\vec{AB}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Show that $\vec{AC} = \frac{1}{2}\vec{AB}$.

[2]

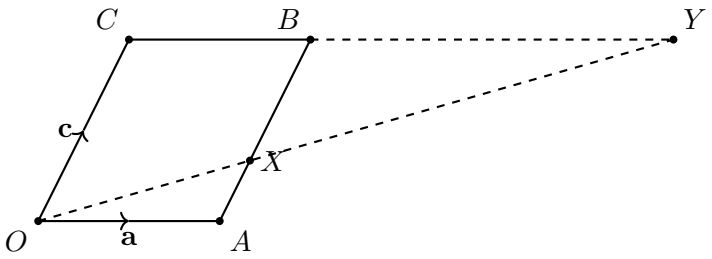
(c) Point C is translated by the vector $\begin{pmatrix} -5 \\ 2 \end{pmatrix}$ to point D.
Find the coordinates of D.

(..... , ...) **[2]**

(d) Work out the vector $\vec{BD}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

17 $OABC$ is a parallelogram.
$\vec{OA} = \mathbf{a}$ and $\vec{OC} = \mathbf{c}$.
X is a point on AB such that $AX : XB = 1 : 2$.
Y is a point on the line CB extended such that $CB : BY = 1 : 2$.



(a) Find $\vec{OX}$ in terms of **a** and **c**.

$\vec{OX} =$ ..... **[2]**

(b) Find $\vec{OY}$ in terms of **a** and **c**.

$\vec{OY} =$ ..... **[2]**

(c) Show that the points O, X and Y lie on a straight line.

[2]

(d) Find the ratio $OX : XY$.

$OX : XY =$ ..... **[1]**

18 The position vectors of points A, B, and C relative to an origin O are given by

$= \begin{pmatrix} 2 \\ -1 \end{pmatrix}$, $= \begin{pmatrix} 6 \\ 7 \end{pmatrix}$, and $= \begin{pmatrix} k+1 \\ 3k-2 \end{pmatrix}$

where k is a constant.

(a) Find $\vec{AB}$ as a column vector.

$\vec{AB} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Given that points A, B, and C are collinear, find the value of k.

$k =$ ..... **[3]**

(c) Using your value of k from part (b), write down $\vec{AC}$ in terms of $\vec{AB}$.

$\vec{AC} =$ ..... **[2]**

19 Given the vectors $\mathbf{u} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.

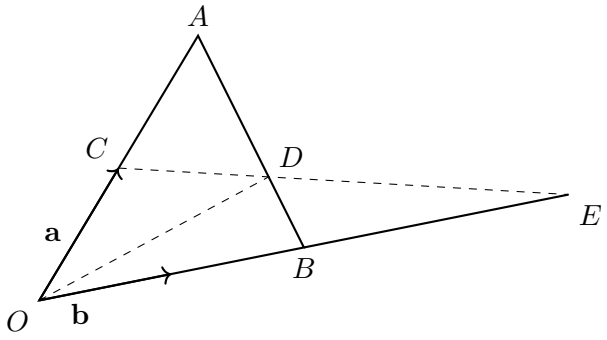
(a) Point P has coordinates $(5, -1)$.
The vector $\vec{PQ} = 2\mathbf{u} - \mathbf{v}$.
Find the coordinates of Q.

(..... , ...) **[3]**

(b) Point R has coordinates $(14, 3)$.
The vector $\vec{PR}$ can be written in the form $h\mathbf{u} + k\mathbf{v}$.
Work out the values of the constants h and k.

$h =$ ..... , $k =$ ..... **[5]**

20 The diagram shows a triangle OAB.
O is the origin, $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$.
C is the midpoint of OA.
D is a point on AB such that $AD : DB = 2 : 1$.
E is a point on OB extended such that $\vec{OE} = 2\mathbf{b}$.



NOT TO SCALE

(a) Find $\vec{AB}$ in terms of **a** and **b**.

$\vec{AB} =$ ..... **[1]**

(b) Find $\vec{OD}$ in terms of **a** and **b**.
Give your answer in its simplest form.

$\vec{OD} =$ ..... **[3]**

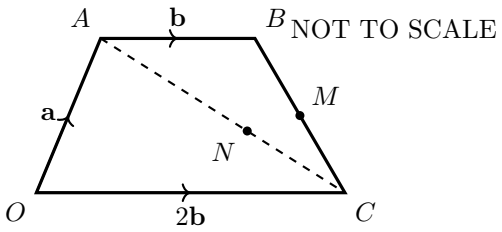
(c) Find $\vec{CE}$ in terms of **a** and **b**.

$\vec{CE} =$ ..... **[2]**

(d) Show that the points C, D and E lie on a straight line.

[2]

21 The diagram shows trapezium $OABC$ with $\vec{OA} = \mathbf{a}$, $\vec{AB} = \mathbf{b}$, and $\vec{OC} = 2\mathbf{b}$.
M is the midpoint of BC, and N is the point on AC such that $AN : NC = 3 : 2$.



(a) Express each of the following vectors in terms of **a** and **b**, giving your answers in their simplest form:

(i) $\vec{OB}$,

$\vec{OB} =$ ..... **[1]**

(ii) $\vec{BC}$,

$\vec{BC} =$ ..... **[2]**

(iii) $\vec{OM}$.

$\vec{OM} =$ ..... **[2]**

(b) Express $\vec{ON}$ in terms of **a** and **b**, giving your answer in its simplest form.

$\vec{ON} =$ ..... **[3]**

(c) Use your answers to part (a)(iii) and part (b) to

(i) explain why O, N, and M lie on a straight line,

[2]

(ii) write down the ratio $ON : NM$ in its simplest form.

$ON : NM =$ ..... **[2]**

1 **(a)** $\begin{pmatrix} 8 \\ -6 \end{pmatrix} + \begin{pmatrix} -2 \\ 5 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 6 \\ -1 \end{pmatrix}$ **[A1]**

(b) $(5, 1)$ **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ **[B1]**

(b) $\begin{pmatrix} 6 \\ 4 \end{pmatrix}$ seen or $2\begin{pmatrix} 3 \\ 2 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 7 \\ 0 \end{pmatrix}$ **[A1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** $\begin{pmatrix} 4 \\ -3 \end{pmatrix} + \begin{pmatrix} -4 \\ 10 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 0 \\ 7 \end{pmatrix}$ **[A1]**

(b) $\begin{pmatrix} 3 \\ -1 \end{pmatrix} + \begin{pmatrix} 4 \\ -3 \end{pmatrix} = \begin{pmatrix} 7 \\ -4 \end{pmatrix}$ [M1]

$(7, -4)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** $3 \times \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 12 \\ -9 \end{pmatrix}$ **[A1]**

(b) $\begin{pmatrix} 4 \\ -3 \end{pmatrix} - \begin{pmatrix} -4 \\ 10 \end{pmatrix}$ or $\begin{pmatrix} 4 - 2(-2) \\ -3 - 2(5) \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 8 \\ -13 \end{pmatrix}$ **[A1]**

(c) $\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -2 \\ 5 \end{pmatrix} = \begin{pmatrix} 6 \\ -1 \end{pmatrix}$ or $(6 - (-2), -1 - 5)$ **[M1]**

$(8, -6)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

5 **(a)** Horizontal component $2 - (-3) = 5$ and vertical component $4 - 1 = 3$ **[M1]**

$\begin{pmatrix} 5 \\ 3 \end{pmatrix}$ **[A1]**

(b) (i) $2\mathbf{q} = \begin{pmatrix} -2 \\ 8 \end{pmatrix}$ **[B1]**

$\begin{pmatrix} 1 \\ 6 \end{pmatrix}$ **[B1]**

(ii) $3\mathbf{p} = \begin{pmatrix} 9 \\ -6 \end{pmatrix}$ **[B1]**

$\begin{pmatrix} 10 \\ -10 \end{pmatrix}$ **[B1]**

(c) $\vec{OE} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} -4 \\ -6 \end{pmatrix} = \begin{pmatrix} -2 \\ -2 \end{pmatrix}$ **[M1]**

$(-2, -2)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

6 (a) (i) $\vec{OA} + \frac{1}{2}\vec{AB} = \mathbf{a} + \frac{1}{2}\mathbf{c}$ **[M1]**

$\mathbf{a} + \frac{1}{2}\mathbf{c}$ **[A1]**

(ii) $\vec{MB} + \vec{BN} = \frac{1}{2}\mathbf{c} - \frac{2}{3}\mathbf{a}$ **[M1]**

$\frac{1}{2}\mathbf{c} - \frac{2}{3}\mathbf{a}$ **[A1]**

(b) $\vec{ON} = \vec{OC} + \vec{CN} = \mathbf{c} + \frac{1}{3}\mathbf{a}$ **[M1]**

$\frac{1}{3}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

7 (a) (i) $\mathbf{b} - \mathbf{a}$ **[B1]**

(ii) $\vec{OP} = \vec{OA} + \frac{1}{4}\vec{AB} = \mathbf{a} + \frac{1}{4}(\mathbf{b} - \mathbf{a})$ **[M1]**

$\frac{3}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}$ **[A1]**

(iii) $\vec{MP} = \vec{MO} + \vec{OP} = -\frac{1}{2}\mathbf{a} + \left(\frac{3}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}\right)$ **[M1]**

$\frac{1}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}$ **[A1]**

(b) $\vec{OC} = \vec{OB} + \vec{BC} = \mathbf{b} + k\mathbf{a}$ **[B1]**

$\mathbf{b} + k\mathbf{a} = m\left(\frac{3}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}\right) = \frac{3m}{4}\mathbf{a} + \frac{m}{4}\mathbf{b}$ **[M1]**

Comparing coefficients of $\mathbf{b}$ gives $\frac{m}{4} = 1$, so $m = 4$ **[A1]**

Comparing coefficients of **a** gives $k = \frac{3(4)}{4} = 3$ **[A1]**

[Back to Question] *[Total Marks: 9]*

8 (a) $\begin{pmatrix} 8 \\ -1 \end{pmatrix}$ **[B1]**

(b) Deduces correct column vectors for **a** and **b**: $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ **[B1]**

Sets up correct simultaneous equations from components: $2p - q = 8$ and $p + 2q = -1$ **[M1]**

Valid method to solve the simultaneous equations (e.g. multiplying first by 2 and adding: $4p - 2q = 16$

5p = 15)$[M1]$

$p = 3$ **[A1]**

$q = -2$ **[A1]**

[Back to Question] *[Total Marks: 5]*

9 (a) Sets up an equation for components, e.g. $2(3) - n = 10$ or $2m - (-2) = 7$ **[M1]**

$n = -4$ **[A1]**

$m = 2.5$ (or $\frac{5}{2}$) **[A1]**

(b) Sets up scalar multiplication equation, e.g. $\begin{pmatrix} k \\ 12 \end{pmatrix} = c \begin{pmatrix} 3 \\ m \end{pmatrix}$ **[M1]**

Finds scalar multiplier $c = \frac{12}{m}$ **[M1]**

$k = \frac{36}{m}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

10 (a) $\begin{pmatrix} 12 \\ -9 \end{pmatrix}$ or $\begin{pmatrix} -4 \\ 10 \end{pmatrix}$ seen **[M1]**

$\begin{pmatrix} 16 \\ -19 \end{pmatrix}$ **[A1]**

(b) $\vec{AB} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ **[M1]**

$(5, 1)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

11 (a) $\begin{pmatrix} 12 \\ -9 \end{pmatrix}$ **[B1]**

(b) $\begin{pmatrix} 4 \\ -3 \end{pmatrix} - \begin{pmatrix} -4 \\ 10 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 8 \\ -13 \end{pmatrix}$ **[A1]**

(c) $(3+4, -1-3)$ **[M1]**

$(7, -4)$ **[A1]**

[Back to Question] *[Total Marks: 5]*

12 (a) $\vec{AB} = \mathbf{b} - \mathbf{a}$ or $\vec{BA} = \mathbf{a} - \mathbf{b}$ **[B1]**

$\vec{OP} = \mathbf{a} + \frac{3}{4}(\mathbf{b} - \mathbf{a})$ or $\vec{OP} = \mathbf{b} + \frac{1}{4}(\mathbf{a} - \mathbf{b})$ **[M1]**

$\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$ **[A1]**

(b) $\vec{MP} = -\frac{1}{2}\mathbf{a} + \vec{OP}$ (oe) **[M1]**

$-\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

13 (a) $\begin{pmatrix} 6 \\ -4 \end{pmatrix} + \begin{pmatrix} x \\ 5 \end{pmatrix} = \begin{pmatrix} -1 \\ y \end{pmatrix}$ **[M1]**

$6 + x = -1 \implies x = -7$ **[A1]**

$-4 + 5 = y \implies y = 1$ **[A1]**

(b) (i) $(4 - 2, 7 + 6)$ **[M1]**

$(2, 13)$ **[A1]**

(ii) $\begin{pmatrix} 2 \\ -6 \end{pmatrix}$ **[B1]**

[Back to Question] *[Total Marks: 6]*

14 (a) Forms a correct equation for the horizontal components: $3(4) - k = 10$ **[M1]**

$k = 2$ **[A1]**

Forms a correct equation for the vertical components: $3(-3) + 2y = 6$

-9 + 2y = 6[M1]

$y = 7.5$ **[A1]**

(b) (i) $\begin{pmatrix} 3 \\ 6 \end{pmatrix}$ **[B1]**

(ii) Finds vector $\vec{QR} = \begin{pmatrix} 6 \\ 12 \end{pmatrix}$ OR $\vec{PR} = \begin{pmatrix} 9 \\ 18 \end{pmatrix}$ **[M1]**

States that $\vec{QR} = 2\vec{PQ}$ (or equivalent scalar multiplier), showing the vectors are parallel, AND mentions that they share a common point Q (or P), concluding they are collinear **[A1]**

[Back to Question] *[Total Marks: 6]*

15 **(a)** $\begin{pmatrix}-3\\2\end{pmatrix}$ **[B1]**

(b) $\vec{BC} = \begin{pmatrix}-9\\6\end{pmatrix}$ **[M1]**

Identifies method to find C, e.g. $(-1,5) + \begin{pmatrix}-9\\6\end{pmatrix}$ **[M1]**

$(-10, 11)$ **[A1]**

(c) $\mathbf{q} = \begin{pmatrix}4\\-1\end{pmatrix} - \begin{pmatrix}-3\\2\end{pmatrix}$ **[M1]**

$\begin{pmatrix}7\\-3\end{pmatrix}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

16 **(a)** $\begin{pmatrix}7\\10\end{pmatrix} - \begin{pmatrix}1\\-2\end{pmatrix}$ **[M1]**

$\begin{pmatrix}6\\12\end{pmatrix}$ **[A1]**

(b) $\vec{AC} = \begin{pmatrix}4-1\\4-(-2)\end{pmatrix} = \begin{pmatrix}3\\6\end{pmatrix}$ **[M1]**

$\frac{1}{2}\vec{AB} = \frac{1}{2}\begin{pmatrix}6\\12\end{pmatrix} = \begin{pmatrix}3\\6\end{pmatrix}$, hence $\vec{AC} = \frac{1}{2}\vec{AB}$ **[A1]**

(c) $(4-5, 4+2)$ **[M1]**

$(-1, 6)$ **[A1]**

(d) $\begin{pmatrix}-1\\6\end{pmatrix} - \begin{pmatrix}7\\10\end{pmatrix}$ or equivalent path method $-\vec{AC} + \vec{CD}$ **[M1]**

$\begin{pmatrix}-8\\-4\end{pmatrix}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

17 **(a)** $\vec{AX} = \frac{1}{3}\mathbf{c}$ **[M1]**

$\mathbf{a} + \frac{1}{3}\mathbf{c}$ **[A1]**

(b) $\vec{BY} = 2\mathbf{a}$ or $\vec{CY} = 3\mathbf{a}$ **[M1]**

$3\mathbf{a} + \mathbf{c}$ **[A1]**

(c) $\vec{OY} = 3\mathbf{a} + \mathbf{c} = 3(\mathbf{a} + \frac{1}{3}\mathbf{c}) = 3\vec{OX}$ **[M1]**

Since $\vec{OY}$ is a scalar multiple of $\vec{OX}$ and they share the common point O, they are collinear. **[A1]**

(d) $1 : 2$ **[B1]**

[Back to Question] *[Total Marks: 7]*

18 **(a)** $\begin{pmatrix}6\\7\end{pmatrix}$ - $\begin{pmatrix}2\\-1\end{pmatrix}$ [M1]

$\begin{pmatrix}4\\8\end{pmatrix}$ **[A1]**

(b) $\vec{AC} = \begin{pmatrix}k-1\\3k-1\end{pmatrix}$ **[B1]**

Gradient or ratio condition: $\frac{3k-1}{k-1} = \frac{8}{4}$ or $\vec{AC} = m\vec{AB}$ **[M1]**

$3k - 1 = 2(k-1) \implies k = -1$ **[A1]**

(c) $\vec{AC} = \begin{pmatrix}-2\\-4\end{pmatrix}$ **[M1]**

$-\frac{1}{2}\vec{AB}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

19 **(a)** $\vec{PQ} = 2\begin{pmatrix}3\\-2\end{pmatrix} - \begin{pmatrix}-1\\4\end{pmatrix} = \begin{pmatrix}7\\-8\end{pmatrix}$ **[M1]**

$\begin{pmatrix}x\\y\end{pmatrix} - \begin{pmatrix}5\\-1\end{pmatrix} = \begin{pmatrix}7\\-8\end{pmatrix}$ or $(5+7, -1-8)$ **[M1]**

$(12, -9)$ **[A1]**

(b) $\vec{PR} = \begin{pmatrix}14-5\\3-(-1)\end{pmatrix} = \begin{pmatrix}9\\4\end{pmatrix}$ **[B1]**

$3h - k = 9$ and $-2h + 4k = 4$ **[M1]**

Rearranging to get $k = 3h - 9$ and substituting: $-2h + 4(3h - 9) = 4$ or equivalent valid method to solve simultaneous equations **[M1]**

$10h = 40 \implies h = 4$ **[A1]**

$k = 3(4) - 9 = 3$ **[A1]**

[Back to Question] *[Total Marks: 8]*

20 **(a)** $-\mathbf{a} + \mathbf{b}$ OR $\mathbf{b} - \mathbf{a}$ **[B1]**

(b) $\vec{OD} = \vec{OA} + \vec{AD}$ OR $\vec{OD} = \vec{OB} + \vec{BD}$ **[M1]**

$\vec{AD} = \frac{2}{3}(\mathbf{b} - \mathbf{a})$ OR $\vec{BD} = \frac{1}{3}(\mathbf{a} - \mathbf{b})$ **[M1]**

$\mathbf{a} + \frac{2}{3}(\mathbf{b} - \mathbf{a}) = \frac{1}{3}\mathbf{a} + \frac{2}{3}\mathbf{b}$ **[A1]**

(c) $\vec{CE} = \vec{CO} + \vec{OE}$ OR $-\frac{1}{2}\mathbf{a} + 2\mathbf{b}$ **[M1]**

$-\frac{1}{2}\mathbf{a} + 2\mathbf{b}$ **[A1]**

(d) Finds $\vec{CD} = \vec{OD} - \vec{OC} = (\frac{1}{3}\mathbf{a} + \frac{2}{3}\mathbf{b}) - \frac{1}{2}\mathbf{a} = -\frac{1}{6}\mathbf{a} + \frac{2}{3}\mathbf{b}$ **[M1]**

Shows $\vec{CE} = 3\vec{CD}$ (or equivalent scalar multiple) and states they share a common point C, concluding they form a straight line. **[A1]**

[Back to Question] *[Total Marks: 8]*

21 **(a)** **(i)** $\mathbf{a} + \mathbf{b}$ **[B1]**

(ii) $\vec{BC} = \vec{BO} + \vec{OC} = -(\mathbf{a} + \mathbf{b}) + 2\mathbf{b}$ **[M1]**

$\mathbf{b} - \mathbf{a}$ **[A1]**

(iii) $\vec{OM} = \vec{OB} + \frac{1}{2}\vec{BC} = (\mathbf{a} + \mathbf{b}) + \frac{1}{2}(\mathbf{b} - \mathbf{a})$ **[M1]**

$\frac{1}{2}\mathbf{a} + \frac{3}{2}\mathbf{b}$ **[A1]**

(b) $\vec{AC} = \vec{AO} + \vec{OC} = 2\mathbf{b} - \mathbf{a}$ **[B1]**

$\vec{ON} = \vec{OA} + \frac{3}{5}\vec{AC} = \mathbf{a} + \frac{3}{5}(2\mathbf{b} - \mathbf{a})$ **[M1]**

$\frac{2}{5}\mathbf{a} + \frac{6}{5}\mathbf{b}$ **[A1]**

(c) **(i)** $\vec{ON} = \frac{2}{5}(\mathbf{a} + 3\mathbf{b})$ and $\vec{OM} = \frac{1}{2}(\mathbf{a} + 3\mathbf{b})$, so $\vec{ON} = \frac{4}{5}\vec{OM}$ **[M1]**

$\vec{ON}$ is a scalar multiple of $\vec{OM}$ and both vectors share the common point O, so O, N, and M are collinear **[A1]**

(ii) $\vec{ON} = \frac{4}{5}\vec{OM}$ gives $ON : OM = 4 : 5$ **[M1]**

$4 : 1$ **[A1]**

[Back to Question] *[Total Marks: 12]*